

12**Ans: A**

Since there is no net transfer of heat energy from one to the other, bodies X and Y are in thermal equilibrium, i.e. same temperature. Their mean KE, given by $\frac{3}{2} kT$ must be the same.

The internal energies and the total kinetic energies of bodies X and Y depends on both their temperature and the number of molecules.

Thermal equilibrium does not depends on thermal conductivity of the material of bodies X and Y.

13**Ans: B**